

# Compressing Arabic Prompts Without Losing the Message

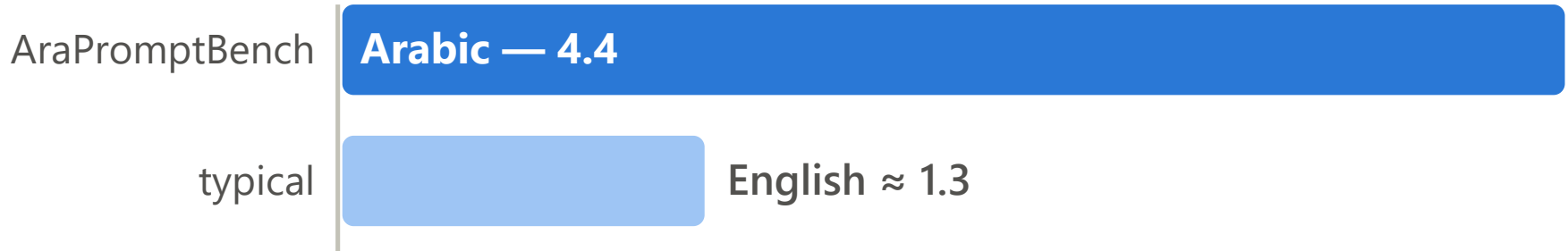
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## 1 Why Arabic prompts need compression

Commercial LLMs charge per token — and their tokenisers, optimised for English, fragment Arabic words into far more tokens than English words of equal meaning. Measured on this project's benchmark, **Arabic costs  $\approx 4.4$  tokens per word versus  $\approx 1.3$  for English**: Arabic users pay roughly three times more per word.

Prompt compression removes low-value tokens before the prompt is sent — but every published method was built and tested on English. This project asks whether those methods transfer to Arabic, and builds **APCS**: a selector that recommends the right compression strategy for each Arabic prompt. The tokeniser tax — c100k tokens per word



Arabic measured across 1,000 benchmark prompts; English figure from Petrov et al. (2023).

## 2 Benchmark & method

**AraPromptBench** — 1,000 Modern Standard Arabic prompts: 850 sourced from six public corpora (CIDAR, XL-Sum, EASC, TyDi-QA, ARCD, Aya) with per-prompt licence tracking, plus 150 synthetic creative briefs; 80/20 development/test split with a contamination-free test set.

- ▶ **Methods compared:** LLMingua (perplexity-based), LLMingua-2 (classifier-based), random deletion, no compression — at keep-rates 0.7 / 0.5 / 0.3.
- ▶ **Two-level evaluation:** token savings (TCR) and semantic fidelity as BERTScore-F1 — between prompt texts, and between the LLM's *answers* to compressed vs original prompts.
- ▶ **Statistics:** paired Wilcoxon, McNemar, seed-42 reproducibility manifests.



Features: token count · structure · fragmentation · morphology · category

## 3 English defaults break on Arabic

LLMingua's default GPT-2 scorer reads Arabic as raw bytes — each letter is two bytes — so deleting "tokens" slices letters in half:

زيادة → زيادة

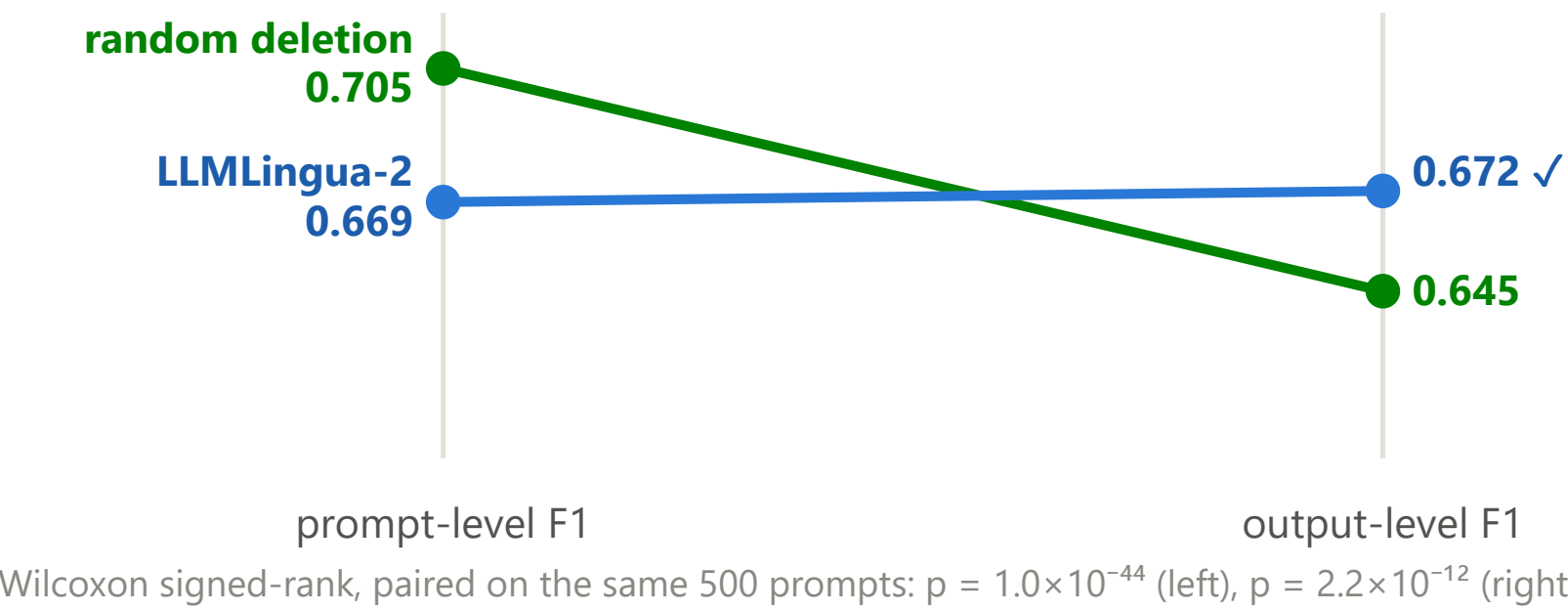
Character corruption in 38 of 45 outputs; crashes on all long prompts; 45% of output words broken. Substituting a multilingual scorer (Qwen2.5-0.5B) fixed corruption — but revealed that LLMingua-1 refuses to compress short prompts at all (96–98% of QA/creative prompts returned untouched). **LLMingua-2's multilingual classifier needed no modification** — hitting every compression target on every category.

## References

Antoun, W., Baly, F. and Hajj, H. (2020) 'AraBERT: Transformer-based model for Arabic language understanding', *Proc. OSACT4*, pp. 9–15. · Chen, L., Zaharia, M. and Zou, J. (2024) 'FrugalGPT', *TMLR*. · Jiang, H. et al. (2023) 'LLMingua: Compressing prompts for accelerated inference of LLMs', *EMNLP 2023*, pp. 13358–13376. · Pan, Z. et al. (2024) 'LLMingua-2: Data distillation for efficient and faithful task-agnostic prompt compression', *Findings of ACL 2024*, pp. 963–981. · Petrov, A., La Malfa, E., Torr, P. and Bibi, A. (2023) 'Language model tokenizers introduce unfairness between languages', *NeurIPS 36*. · Habash, N.Y. (2010) *Introduction to Arabic Natural Language Processing*. Morgan & Claypool.

## 4 Measure answers, not prompts — the ranking flips

Judged by prompt-text similarity, random deletion looks *better* than intelligent compression. Judged by whether the LLM's *answer* is preserved, the ranking inverts — both with  $p < 10^{-11}$ . Studies using the cheap metric would crown the wrong method. **BERTScore-F1 at keep-rate 0.5 (AraBERT scorer, n = 500 pilot)**



Wilcoxon signed-rank, paired on the same 500 prompts:  $p = 1.0 \times 10^{-44}$  (left),  $p = 2.2 \times 10^{-12}$  (right).

13%

of identical prompts give an identical answer twice (temp 0)

31%

of identical-prompt answer pairs fail the planned 0.85 threshold

0.70

re-derived threshold: fails only 0.4% of pure noise

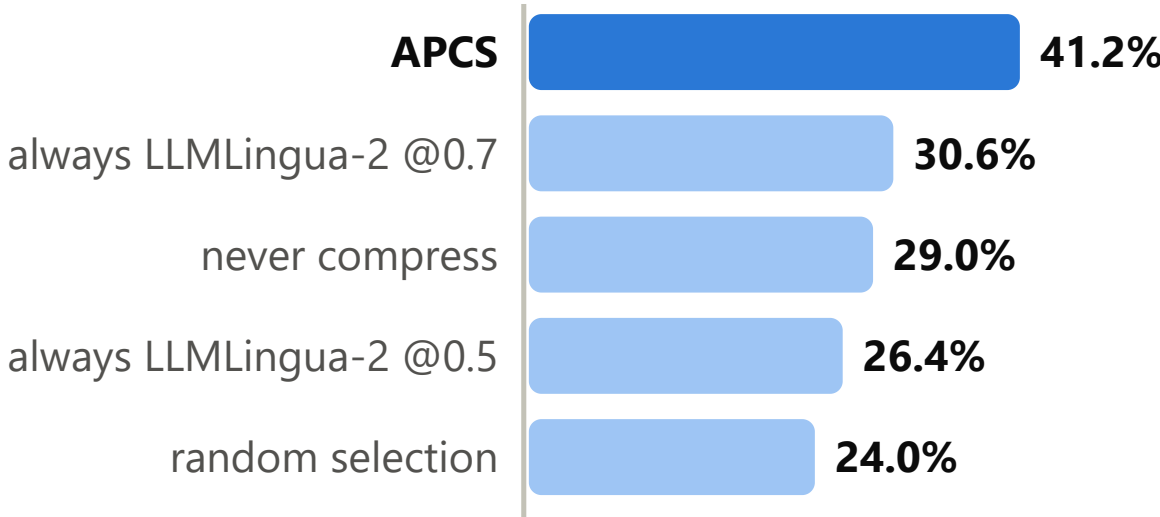
The fidelity threshold was recalibrated against the LLM's measured noise ceiling — a methodological contribution applicable beyond Arabic.

## 5 APCS — the right strategy per prompt

No single choice is optimal for even a third of prompts. APCS extracts lightweight features and applies an empirically calibrated, fully explainable rule:

if `token_count < 120` → do not compress  
otherwise → LLMingua-2 @ keep-rate 0.5

Optimal-strategy selection accuracy (pilot, n = 500)



APCS beats every fixed baseline (McNemar exact,  $p \leq 6 \times 10^{-3}$ ) and captures 80% of the theoretical-maximum token savings at near-optimal fidelity — using one explainable rule. Total API cost of all experiments: under \$2.

## 6 Conclusions & next steps

- ▶ English-default compression does not transfer to Arabic; multilingual classifier-based compression (LLMingua-2) does.
- ▶ Prompt length — not morphology — drives compression tolerance; QA prompts suffer most once the LLM's noise ceiling is accounted for.
- ▶ Adaptive selection beats any fixed strategy: the premise of APCS holds for Arabic.
- ▶ **Next:** final held-out evaluation on the 1,000-prompt benchmark with a current-generation LLM, and public release of AraPromptBench and the APCS package.